CHAIN OF CUSTODY FORM (Original Stays With Samples)	US EPA Experimental Stream Facility 1003 US Highway 50, Milford, Ohio 45150 Phone # 513-965-4805 FAX # 513-965-4806	Originator:	
		Ship Date: 05 22 2018	Contact:
		Destination: Cincinnati	Phone #:
		Recipient: Sam Smith	E-Mail:
		Phone #:	

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180521	14:10:00				1	DI	Field BLK01		TNH4, TRP, TNO3, TNO2, TN, TP
20180521	14:13:00				1	DI	Field BLK02		"
20180521	14:15:00				1	DI	BTL BLK01		"
20180521	14:17:00				1	DI	BTL BLK02		"
20180522	08:28:00	FL	42	0.1	1	SW	UKN		"
20180522	08:46:00	FL	44	0.1	1	SW	UKN		"
20180522	09:30:00	FL	43	0.1	1	SW	UKN		"
20180522	09:56:00	FL	41	0.1	1	SW	UKN		"
20180522	09:56:00	FL	41	0.1	2	SW	DUP		"
20180522	10:41:00	FL	04	0.1	1	SW	UKN		"
20180522	11:05:00	FL	03	0.1	1	SW	UKN		"
20180522	11:19:00	FL	14	0.1	1	SW	UKN		"
20180522	11:39:00	FL	15	0.1	1	SW	UKN		"
20180522	12:23:00	FL	07	0.1	1	SW	UKN		"
20180522	12:44:00	FL	01	0.1	1	SW	UKN		"
20180522	13:28:00	FL	05	0.1	1	SW	UKN		"
20180522	13:42:00	FL	02	0.1	1	SW	UKN		"
20180522	13:42:00	FL	02	0.1	1	SW	DUP		"
20180523	11:15:00				1	DI	HOUSE DI		"
20180521	14:10:00				1	DI	Field BLK01	2M HCl	TOC
20180521	14:13:00				1	DI	Field BLK02	"	TOC
20180521	14:15:00				1	DI	BTL BLK01	"	TOC
20180521	14:17:00				1	DI	BTL BLK02	"	TOC
20180522	08:28:00	FL	42	0.1	1	SW	UKN	"	TOC
20180522	08:46:00	FL	44	0.1	1	SW	UKN	"	TOC
20180522	09:30:00	FL	43	0.1	1	SW	UKN	"	TOC
20180522	09:56:00	FL	41	0.1	1	SW	UKN	"	TOC
20180522	09:56:00	FL	41	0.1	2	SW	DUP	"	TOC
20180522	10:41:00	FL	04	0.1	1	SW	UKN	"	TOC

Relinquished By:	Date:	Time:	Received By:	Date:		Time:

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:	
Ship Date: 05 22 2018	Contact:
Destination: Cincinnati	Phone #:
Recipient: Sam Smith	E-Mail:
Phone #:	

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180522	11:05:00	FL	03	0.1	1	SW	UKN	2M HCl	TOC
20180522	11:19:00	FL	14	0.1	1	SW	UKN	"	TOC
20180522	11:39:00	FL	15	0.1	1	SW	UKN	"	TOC
20180522	12:23:00	FL	07	0.1	1	SW	UKN	"	TOC
20180522	12:44:00	FL	01	0.1	1	SW	UKN	"	TOC
20180522	13:28:00	FL	05	0.1	1	SW	UKN	"	TOC
20180522	13:42:00	FL	02	0.1	1	SW	UKN	"	TOC
20180522	13:42:00	FL	02	0.1	2	SW	DUP	"	TOC
20180523	11:15:00				1	DI	HOUSE DI	"	TOC

Relinquished By:	Date:	Time:	Received By:	Date:		Time:

Di-ionized

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:		
Ship Date:	05222018	Contact:
Destination:	Cincinnati	Phone #:
Recipient:	Sam Smith	E-Mail:
Phone #:		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)	
20180521	14:10:00				1	DI	FLD BLK d	chlorophyll	392.06	
20180521	14:13:00				1	DI	FLD BLK 02	"	376.49	
20180522	08:28:00	FL	42	0.1	1	SW	UKN	"	141.39	
20180522	08:46:00	FL	44	0.1	1	SW	UKN	"	168.01	
20180522	09:30:00	FL	43	0.1	1	SW	UKN	"	198.99	
20180522	09:56:00	FL	41	0.1	1	SW	UKN	"	187.76	
20180522	09:56:00	FL	41	0.1	1	SW	DUP	"	187.85	
20180522	10:41:00	FL	04	0.1	1	SW	UKN	"	220.7	
20180522	11:05:00	FL	03	0.1	1	SW	UKN	"	208.52	
20180522	11:19:00	FL	14	0.1	1	SW	UKN	"	231.16	
20180522	11:39:00	FL	15	0.1	1	SW	UKN	"	248.89	
20180522	12:23:00	FL	07	0.1	1	SW	UKN	"	281.67	
20180522	12:44:00	FL	01	0.1	1	SW	UKN	"	361.67	
20180522	13:28:00	FL	05	0.1	1	SW	UKN	"	328.7	
20180522	13:42:00	FL	02	0.1	1	SW	UKN	"	302.63	
20180522	13:42:00	FL	02	0.1	1	SW	DUP	"	310.3	
						surface water				

	Relinquished By:	Date:	Time:	Received By:			

GAS TRAP DATA SHEET

Lake:			5/22/18	5/23/18		05232018.
Site	Lat	Long	Deployment time/date	Retrieval time/date	Volume recovered (mL)	Exetainer codes
53	36.069178	78.794701	8:20	8:20	15	Fl.53.BT.1
42	36.065443	78.795194	8:28	8:34	0	
50	36.068120	78.786094	8:38	8:45	7	
44	36.072778	78.779461	8:46	8:57	2	
52	36.079507	78.763176	9:01	9:11	6	
56	36.072872	78.766419	9:11	9:23	4	
47	36.068272	78.766506	9:20	9:37	6	
43	36.061527	78.773301	9:30	9:47	1	
54	36.057859	78.768062	9:40	10:10	4	
55	36.053040	78.761341	9:48	10:21	9	
41	36.055463	78.758776	9:56	10:30	1	
49	36.051530	78.756689	10:04	10:40	25 25 3	Fl.49.BT.1 Fl.49.BT.2
57	36.048318	77.752465	10:12	10:54	— 0	Valve open
45	36.044025	78.758174	10:19	11:05	2	
10	36.039890	78.756301	10:29	11:13	4	
4	36.012277	78.761039	10:41	11:25	0	
8	36.019133	78.748236	10:55	11:36	9	
3	36.033452	78.737681	11:05	11:47	12.5	
14	36.059732	78.717159	11:19	12:04	0	
15	36.022950	78.714637	11:39	12:23	1	
11	36.019280	78.716904	11:31	12:47	1	
7	36.005415	78.667762	12:23	13:22	0.5	
1	36.022085	78.634336	12:44	13:41	1.5	
21	36.008950	78.628261	12:51	13:53	0	
13	36.003603	78.646435	13:02	14:05	0	
12	35.965463	78.634056	13:18	14:26	0.1	
5	35.970178	78.621316	13:28	14:38	25 25	Fl.5.BT.1 Fl.5.BT.2
7	35.976308	78.589002	13:42	14:59	13	
	956608	78.590357	13:57	15 1	0.5	
24	35.942107	78.588142	14:10	15:28	15	Fl.24.BT.1

DISSOLVED GAS AND AIR DATA SHEET

Lake:	F41/5				Date:	5/23/18		
¥Site	Dissolved gas Exetainer codes 05232018. 05232018.		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes 05232018.			BP (mm Hg)
42	F1.42.DG.1	F1.42.DG.2	115/25	26.2				
44	F1.44.DG.1	F1.42.DG.2	11	26.3				
43	F1.43.DG.1	F1.43.DG.2	11	27.1				
41	F1.41.DG.1	F1.41.DG.2	11	29.1	F1.41.AA.1	F1.41.AA.2	F1.41.AA.3	F1.41.AA.4 F1.AA.41.5
4	F1.4.DG.1	F1.4.DG.2	11	28.9				
3	F1.3.DG.1	F1.3.DG.2	11	30.1				
15	F1.15.DG.1	F1.15.DG.2	11	30				
14	F1.14.DG.1	F1.14.DG.2	11	29.1				
7	F1.7.DG.1	F1.7.DG.2	11	30.1				
1	F1.1.DG.1	F1.1.DG.2	11	30.9				
5	F1.5.DG.1	F1.5.DG.2	11	31.5				
2	F1.2.DG.1	F1.2.DG.2	11	31.0	F1.2.AA.1	F1.2.AA.2	F1.2.AA.3	F1.2.AA.4 F1.2.AA.5

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Water Vol/Air Vol :: 115ml/25ml

Goal is to inject 20mL of gas into exetainer

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	FALLS LAKE
Sample Date:	5/23/18
GGA file	339 . 2018-05-23_f0000.txt
gps file name	

DEPLOYMENT DATA

BADEIF=14:33:00 LGR=15:31:27 clocks

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	Water Temperature (C)	CH_4 and CO_2 profile checked
53	8:20:41	N	✓	60	24.41	✓
42	8:33:11	N	✓	50	24.6	✓
50	8:43:57	N	✓	50	24.5	✓
44	8:57:28	Y	✓	50	24.4	✓
52 ~~56~~	9:11:45	N	✓	50	27.1	✓
56	9:24:00	Y	—	—	26.9	start over
56	9:29:26	N	✓	50	26.9	✓
47	9:37:27	N	✓	50	26.9	✓
43	9:47:35	N	✓	50	26.1	✓
54	10:10:10	N	✓	50	26.9	✓
55	10:21:43	N	✓	45	26.6	✓
41	10:29:45	Y	✓	45	27.0	✓
49	10:40:30	Y	✓	45	26.9	✓
57	10:54:08	N	✓	50	27.8	✓
45	11:04:57	N	✓	50	26.4	✓
10	11:13:45	N	✓	50	28.0	✓
4	11:25:45	N	✓	50	26.9	✓
8	11:36:39	N	✓	45	28.9	✓
3	11:47:35	Y	✓	50	29.2	✓
14	12:04:30	N	✓	45	28.3	✓
15	12:23:16	N	✓	45	29.0	✓
11	12:47:00	N	✓	45	29.0	✓
7	13:22:40	N	✓	45	29.0	✓
1	13:40:50	N	✓	45	29.2	✓
21	13:53:00	N	✓	45	30.2	✓
13	14:06:50	N	✓	45	29.6	✓
12	14:26:10	N	✓	45	29.8	✓
5	14:38:15	N	✓	45	29.9	strange 2 regional profile waves?
5	14:43:02	N	✓	45	29.9	✓
2	14:59:05	N	✓	40	30.7	✓
16	15:14:30	Y	✓	40	—	bubble
16	15:16:30	N	✓	40	30.7	✓
24	15:28:23	✓	✓	40	30.1	✓

SONDE DATA SHEET

Lake: Sample Date:

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
53	6.9 ft	0.1	24.21	8.71	104.1	0.077	7.97	13.0	23.9	
42	2.1 m	0.1	23.93	6.44	75.2	0.109	7.79	14.2	30.9	
50	1.9	0.1	23.84	8.10	95.7	0.083	7.66	14.8	24.4	
44	2.4	0.1	24.80	8.71	105.5	0.070	7.61	18.5	23.7	
44	2.4	2.4	22.08	[illegible]	43.7	0.070	7.48	8.1	~~22~~ 57.1	
52	2.6	0.1	25.83	7.56	91.9	0.102	7.34	13.0	10.0	
52	2.6	2.3	22.83	3.65	42.3	0.079	6.88	9.2	9.7	
56	3.4	0.1	24.02	8.31	102.5	0.102	7.28	13.0	10.0	
56	3.4	3.3	22.94	55.6 ↔ 4.81		0.075	6.86	7.8	28.7	
47	3.0	0.1	25.93	8.88	109.2	0.099	7.33	15.5	11.0	
47	3.0	2.7	22.4	4.10	47.1	0.073	7.415	8.3	39.6	
43	3.0	0.1	25.31	9.56	116.6	0.086	7.35	19.5	15.2	
43	3.0	2.7	22.46	4.75	53.3	0.072	7.40	8.0	51.3	
54	3.0	0.1	24.95	9.32	113.1	0.084	7.29	21.2	18.3	
54	3.0	2.7	22.90	5.00	57.0	0.078	7.37	9.8	64.2	
55	3.3	0.1	24.77	9.12	109.6	0.081	7.29	20.0	17.7	
55	3.3	3.2	21.88	~~4.00~~ 4.83	45.1	0.076	7.28	10.8	30.6	
41	2.0	0.1	25.74	9.98	122.6	0.081	7.26	20.8	14.0	
41	2.0	1.7	24.72	8.0	95.0	0.080	7.40	14.2	21.7	
49	3.7	0.1	25.16	9.64	117.8	0.080	7.36	16.6	16.5	
49	3.7	3.6	21.77	3.74	42.5	0.077	7.37	9.8	55.0	
57	3.3	0.1	25.95	9.8	121.3	0.083	7.27	17.1	11.0	
57	3.3	3.2	22.22	4.7	53.0	0.076	7.41	10.6	65.0	
45	3.5	0.1	25.67	10.2	126.3	0.083	7.43	19.8	12.9	
45	3.5	3.4	22.35	5.21	59.8	0.075	7.58	7.9	30.3	
10	1.8	0.1	25.41	9.48	115.8	0.084	7.53	22.8	13.4	
4	2.1	0.1	26.02	7.46	92.8	0.095	7.47	11.4	11.7	
8	5.0	0.1	25.61	8.95	109.5	0 '3	7.55	15.0	9.4	
8	5.0	4.7	22.58	2.68	31.5	0.085	7.49	7.8	27.3	

Date 5/21/18

Cal cylinder serial # CC49885 7

Dilution system serial # A 20808

LGR file

JS9_2018-05-21_f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.027	0.86	15:22:00	15:25:20
9	60.0	2.0	20.2	1.83	19.3	15:25:20	15:28:10
8.5	60.0	2.1	21.4	1.92	20.3	15:28:10	15:30:40
8	60.0	2.2	22.7	2.03	21.4	15:30:40	15:32:50
7.5	60.0	2.4	24.2	2.16	22.7	15:32:50	15:34:50
7	60.0	2.6	25.9	2.31	24.0	15:34:50	15:36:58
6.5	60.0	2.7	27.9	2.48	26.0	15:36:58	15:39:10
6	60.0	3.0	30.2	2.69	28.2	15:39:10	15:41:19
5.5	60.0	3.2	32.9	2.92	30.4	15:41:19	15:43:33
5	60.0	3.6	36.1	3.22	33.5	15:43:33	15:45:30

Signature _____

Date 5/21/18
Cal cylinder serial # CC163816
Dilution system serial # A20808
LGR file 559 - 2018 - 15-21_f0000.txt
Sample MFC gas correction factor 1.43

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.023	1.24	14:53:40	14:55:50
7	70.0	28.3	283.2	25.52	252.4	14:56:10	14:59:00
6.5	70.0	30.5	304.7	27.51	280.3	14:59:00	15:02:07
6	70.0	33.0	329.8	29.82	303.7	15:02:07	15:05:15
5.5	70.0	35.9	359.4	32.53	332.70	15:05:15	15:08:45
5	70.0	39.5	394.9	35.73	367.2	15:08:45	15:12:11
4.5	70.0	43.8	438.1	39.50	408.5	15:12:11	15:14:38
4	70.0	49.2	491.9	44.23	457.6	15:14:38	15:16:54
3.5	70.0	56.1	560.8	50.10	521.6	15:16:54	15:19:00
3	70.0	65.2	652.1	58.00	610.2	15:19:00	15:21:11

Signature _____

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD 16 **DATE:** 5/21/18 **TECHNICIAN:** JTW

DATE / TIME OK? YES / (NO)

BATTERY VOLTS: 12.4 (Change if < 10.0 v) **CHANGED BATTERIES?** YES / (NO)

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 754.6 mm Hg

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

_____ _____ _____ _____ _____

_____ _____ _____ _____ _____

DO MEMBRANE CHANGED? YES / (NO) **PROBE RE-SURFACED?** YES / (NO) **DATE/TIME:** _____

[~97] **INITIAL:** 98.0 **CALIBRATED:** 99.3 **POST-DEPLOYMENT:** 100.8

DO CHARGE ~~[50 ± 25]~~: _____ **DO GAIN** [1.0 ± 0.35]: _____

GAIN 1.06667

DEPTH: "Pressure" ~~[m]~~ ft **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: 0.1002?

[0] **INITIAL:** -0.162 **CALIBRATED:** 0.0 **POST-DEPLOYMENT:** 0.220

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: 4.96146 (pre)

1.413 @ 25°C [10] **INITIAL:** 1.402 **CALIBRATED:** 1.413 **POST-DEPLOYMENT:** 1.406

Temp = 22.93 Temp = 22.86

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** -0.2 **CALIBRATED:** 0.0 **POST-DEPLOYMENT:** -0.3

126 ~~[1000]~~ **INITIAL:** 126 **CALIBRATED:** 124.3 **POST-DEPLOYMENT:** 123.1

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** 0.1 **CALIBRATED:** _____ **POST-DEPLOYMENT:** -0.1

22/18 Site 2 Section Site 41 ~~[113]~~ **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 3.2 Temp 10.59 6/12/18

7.9 Temp 9.99 6/12/18

---

pH: "pH"

[7] **INITIAL:** 7.15 **CALIBRATED:** 7.00 **POST-DEPLOYMENT:** 7.00 **mV** [0 ± 40]: -45.8 Pre

[10] **INITIAL:** 10.0 **CALIBRATED:** 9.96 **POST-DEPLOYMENT:** 9.99 **mV** [-177 ± 40]: -217.0

ORP: "ORP mV" **ORP mV OFFSET [0 ± 100]:** _____

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